Brad Cao

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Education

Columbia University New York, NY
B.A. in Mathematics & Computer Science *Expected May 2030*

- **Relevant Coursework:** Natural Language Processing, Probability Theory, Honors Mathematics A & B, Data Structures in Java
- **Activities:** Columbia AI Alignment Technical Fellowship, Columbia Alpha Partners (Global Markets Division), Blockchain @ Columbia (Investment Fund), Columbia Poker Club

Experience

Research Engineering Intern, Sampura Research · *Remote* Oct. 2026 – Present

- Offered this internship following my SPAR work; work with Rishub Jain and Joshua Jacob (ex-Google DeepMind)
- Onboarded 3 datasets end to end (MASK, JuICE, CRJudgeBench; 1,672 items): HF-pinned transforms, specs, DVC-tracked gate runs, and a JuICE rebuild that counts only human-found errors; gpt-6-luna (K=5) scores 63–73% balanced
- Ran headroom pilots and label audits on 40+ datasets, rejecting most as saturated (gpt-6-luna $\geq$85%) or unreliable; screened $\sim$150 more and surfaced JuICE for the empty cultural-bias category
- Reviewed teammates' onboarding PRs before the v1 freeze; on one, re-derived all 208 gold labels with an independent solver and traced the judge's chance-level score to an always-"no" collapse

Research Fellow, Supervised Program for Alignment Research (SPAR) · *Remote* Sep. 2026 – Present

- Audit datasets for a 100+ dataset leaderboard rating human, AI, and hybrid judges on deception, reward hacking, and bias
- Found label errors that sink published benchmarks: 61% of SYCON-Bench's CREPE answer key wrong or debatable (122/200), $\sim$20% of Sycophantic Anchors labels flipped both ways (39/200), and $\sim$25% of JudgeStressTest's "benign" labels wrong
- Rebuilt MASK as a lie-detection test with a Gemini 3.1 Pro responder (358 lies, 144 verified-honest of 502 items); closed-book, gpt-6-luna catches 99.4% of lies but clears only 25.7% of honest answers

Data Science Intern, Mercor · *San Francisco, CA* Apr. 2026 – Sep. 2026

- Designed 10+ rubric-based RLHF scoring taxonomies that turn expert judgment into consistent LLM reward signal
- Wrote pass/fail criteria with worked edge cases and built QA workflows to raise inter-annotator agreement

Research Assistant, Duke University · *Durham, NC* Apr. 2026 – Sep. 2026

- Built the full LLM pipeline for a cable-news disparagement study (3,543 transcripts, 18.7M words), validated against human-coded samples; found Fox disparages the media at 10× CNN's rate

Leadership Seminar Scholar, Notre Dame · *South Bend, IN* Jun. 2025 – Aug. 2025

- Won a full-ride scholarship to the "Inequality in America" seminar; reviewed 15+ studies on the pharmaceutical industry's effect on health policy and presented at a symposium

Research Intern, University of Maryland, Baltimore · *Baltimore, MD* Jun. 2023 – Sep. 2025

- Predicted serious outcomes for ADHD patients on stimulants from 10 years of FDA FAERS reports (best model 0.75 AUC, 88% recall); age, sex, and drug type were the key risk factors
- First-authored 2 studies presented at the ASCPT Annual Meetings (2024, 2025); the 2025 poster won the Top Poster Award

Projects

Threshold | *Python, NumPy, SciPy, PyTorch* | github.com/bradca0/threshold Solo project

- Reproduced double descent on MNIST: test error peaks at 88.8% at the interpolation threshold (90% of it from variance) and falls to 12.3% with tuned ridge
- Packaged as a reproducible Python library: 85 tests, 94% coverage, `mypy --strict`

MyCureHope | *Django, PostgreSQL, JavaScript, TailwindCSS* | mycurehope.org Co-developer

- Open-source database of treatments, doctors, and hospitals used by 33,000+ users in 78 countries; built search, deployed the stack

Publications

Cao, B., Lee, S., Naga, A., Gobburu, J., Dunn, A. "Leveraging Supervised Machine Learning to Identify Predictors of Serious Clinical Outcome with Stimulant Treatment from Real-World Data." *ASCPT Annual Meeting*, 2025.

Cao, B., Gupta, P., Dunn, A. "Predictive Modeling of Tacrolimus Dose Requirements and Transplantation Outcomes in Lung Transplant Recipients on Systemic Azole Antifungals." *ASCPT Annual Meeting*, 2024.

Technical Skills & Interests

Alignment & Evals: LLM-as-Judge Evaluation, Headroom Testing, Label Auditing, Dataset Curation, RLHF Rubric Design, Mechanistic Interpretability (TransformerLens, GPT-2 Small IOI Circuit)
Languages & Tools: Python, Java, R, SQL (PostgreSQL), JavaScript; PyTorch, scikit-learn, NumPy, pandas, Django, Linux, DVC, GCP
Interests: Fencing (USA Fencing First Team All-American), Competitive Programming (USACO), History Olympiads, Open-Source